
Neural Networks for NLP - Sheet 5

20.01.2025

Deadline: 03.02.2026 - 13:15, on Olat as **.pdf/.html/.ipynb** + **.txt** and/or after the lecture

Task 1: Pre-training and Fine-tuning

12 Points

- (a) **(2 points)** Is there any advantage to using transfer learning when you could just learn from given examples only? Explain with a suitable example.

Imagine you are trying to classify the sentiments of movie reviews as positive or negative. You apply a machine learning method for the task and note the accuracy A_1 . You also use a well-known pre-trained NLP model and get the accuracy A_2 . Surprisingly, you find that $A_1 < A_2$. Could this happen with a real dataset and an established NLP model like BERT? If so, explain in not more than 2 sentences what could have led to such undesirable results.

- (b) **(10 points)** In this exercise, you are assigned the task of classifying whether a given English sentence is grammatically correct or not. You will be working with the *Corpus of Linguistic Acceptability (CoLA)* dataset that contains 10657 sentences from 23 linguistics publications, expertly annotated for acceptability (based on grammar) by their original authors.

Your task is to preprocess this dataset, download a pretrained BERT model, and fine-tune it for this task. Complete the missing pieces of code in the attached Jupyter Notebook and run it to show your results.

Task 2: Variational autoencoder based Topic Model

12 Points

For the following task, please download the NeurIPS dataset [here](#).

- (a) **(3 points)** Use the raw NeurIPS dataset covering the time period of 1987-2019 with 9,680 documents. Get ten years of data from the downloaded file and apply preprocessing steps (such as tokenization and removing stopwords).
- (b) **(6 points)** In the Jupyter notebook, you will find code for the Gaussian variational autoencoder. Based on this code, implement a variational autoencoder using the Dirichlet distribution as a prior on the latent variables.
- (c) **(3 points)** Train a topic model with 25 topics. Save the ten top words for each topic into a file. Train models with three different values of $\alpha \in \{0.1, 1.0, 10.0\}$.